

Simple Missing Data Treatments

Utrecht University Winter School: Missing Data in R



**Utrecht
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Outline

Bad Methods

- Deletion-Based Methods

- Deterministic Imputation Methods

OK Methods

Comparisons



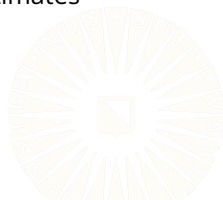
Bad Methods (These almost never work)

Listwise Deletion (Complete Case Analysis)

- Use only complete observations for the analysis
 - Very wasteful (can throw out lots of useful data)
 - Loss of statistical power

Pairwise Deletion (Available Case Analysis)

- Use only complete pairs of observations for analysis
 - Different samples sizes for different parameter estimates
 - Can cause computational issues



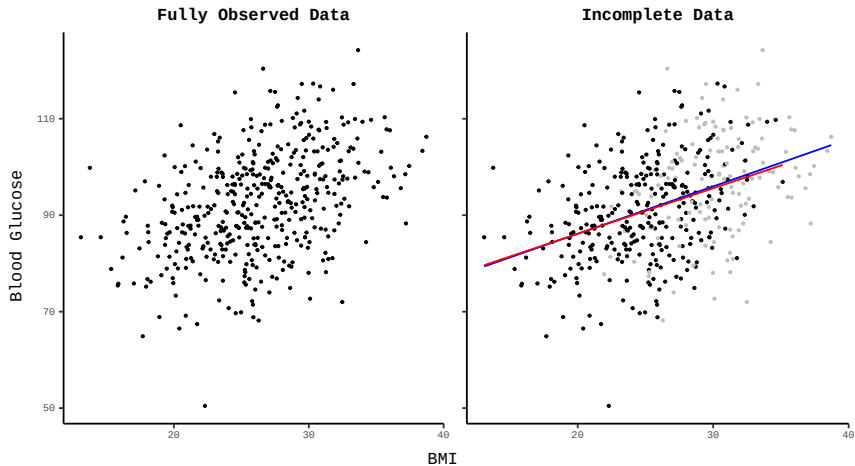
Example

```
## Read in some example data:
dat0 <- dat1 <- readRDS(paste0(dataDir, "diabetes_norm.rds"))

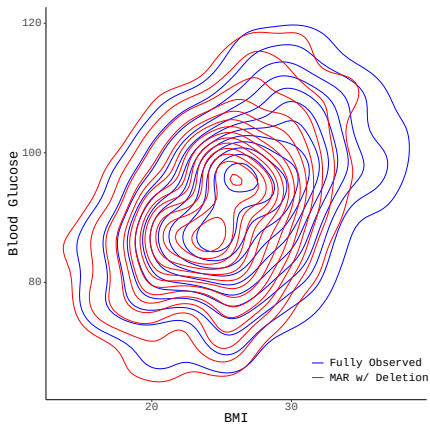
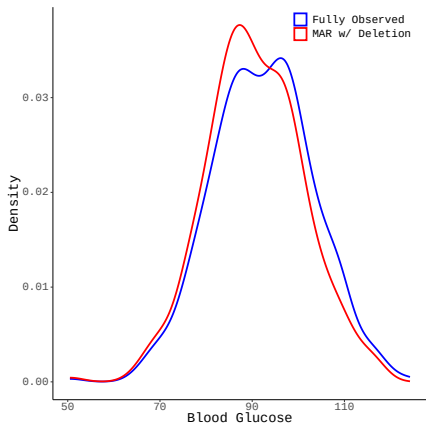
## Simulated missingness based on 'bmi':
m <- simLinearMissingness(data = dat1,
                           pm    = 0.30,
                           preds = "bmi",
                           auc   = 0.85)$r

## Impose missing on 'glu' according to the missingness above:
dat1[m, "glu"] <- NA
```

Example



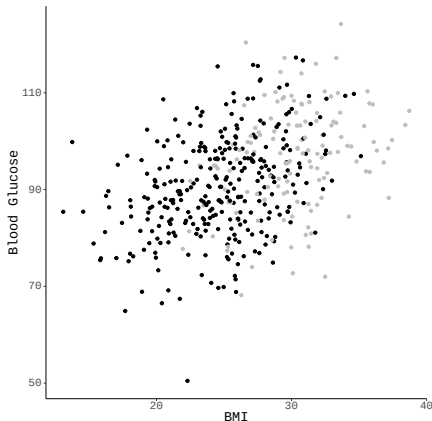
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Bad Methods (These almost never work)

(Unconditional) Mean Substitution

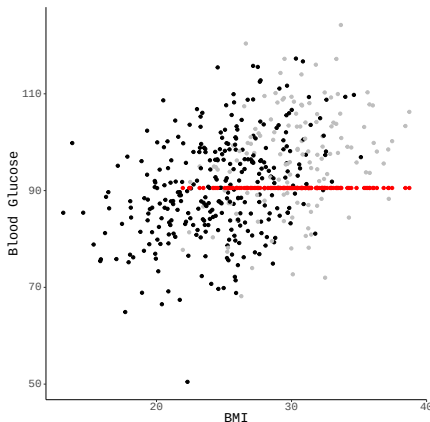
- Replace Y_{mis} with $\bar{Y}_{obs}$
 - Negatively biases regression slopes and correlations
 - Attenuates measures of linear association



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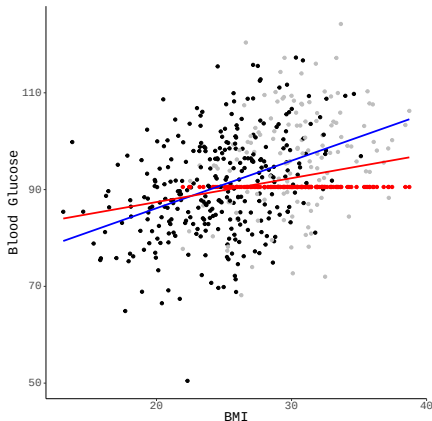
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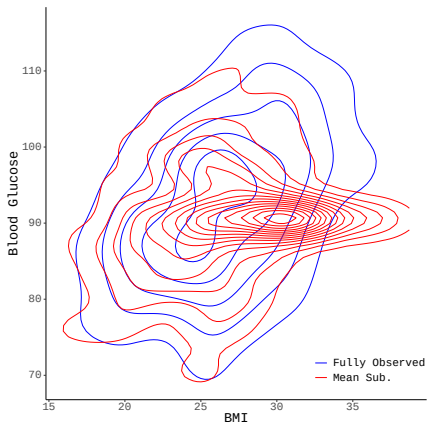
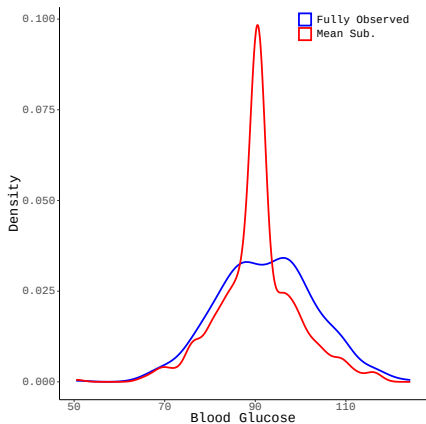
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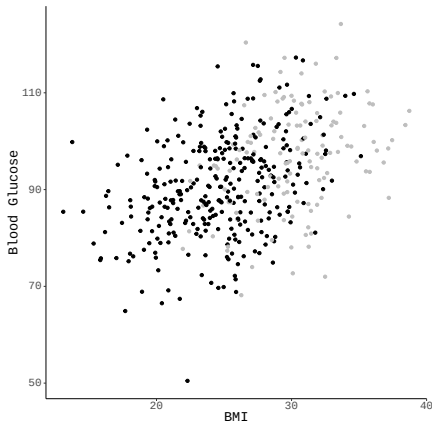
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Bad Methods (These almost never work)

Deterministic Regression Imputation (Conditional Mean Substitution)

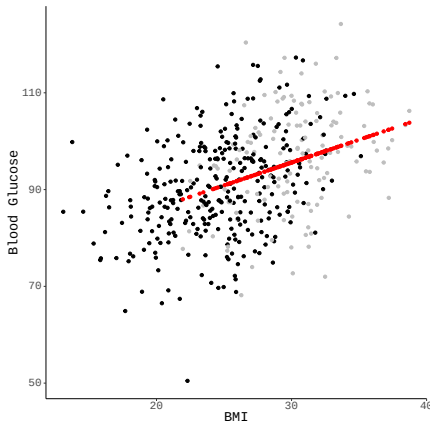
- Replace Y_{mis} with $\hat{Y}_{mis}$ from some regression equation
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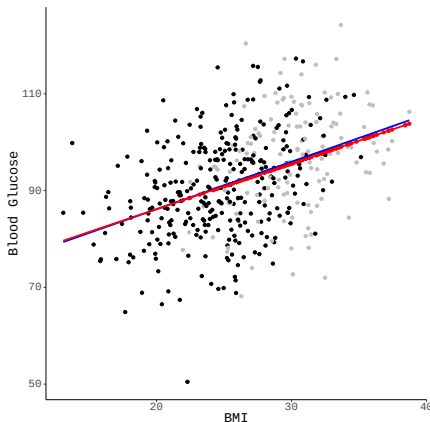
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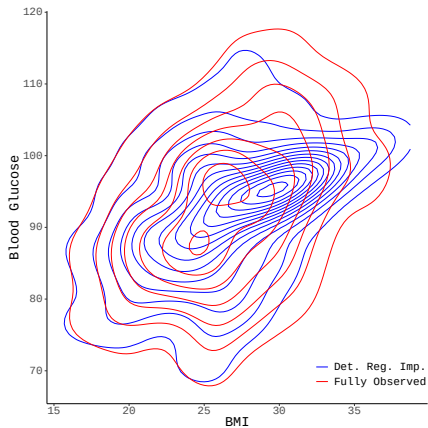
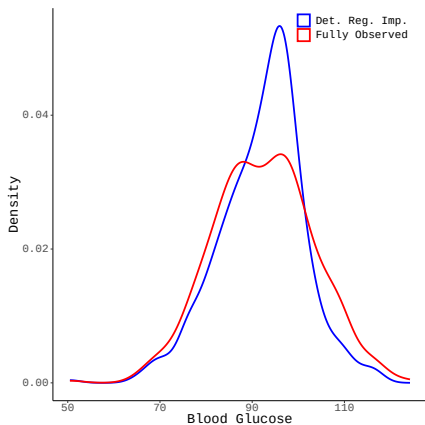
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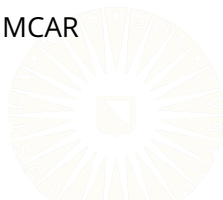
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General Issues with Deletion-Based Methods

- Biased parameter estimates unless data are MCAR
- Generalizability issues

General Issues with Simple Single Imputation Methods

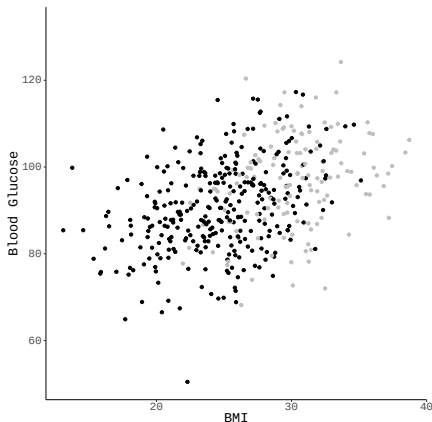
- Biased parameter estimates even when data are MCAR
- Attenuates variability in any treated variables



OK Methods (These sometimes work)

Stochastic Regression Imputation

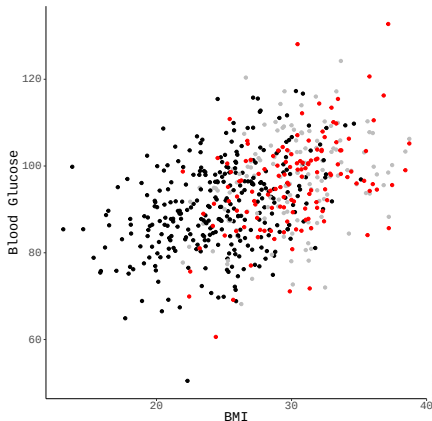
- Fill Y_{mis} with $\hat{Y}_{mis}$ plus some random noise.
 - Produces unbiased parameter estimates and predictions
 - Computationally efficient
 - Attenuates standard errors
 - Makes CIs and prediction intervals too narrow



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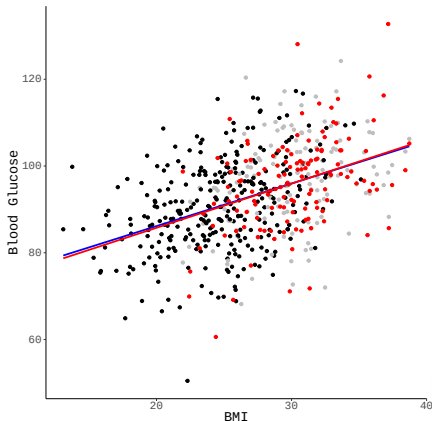
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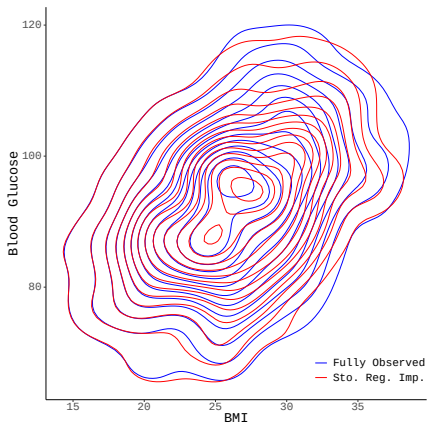
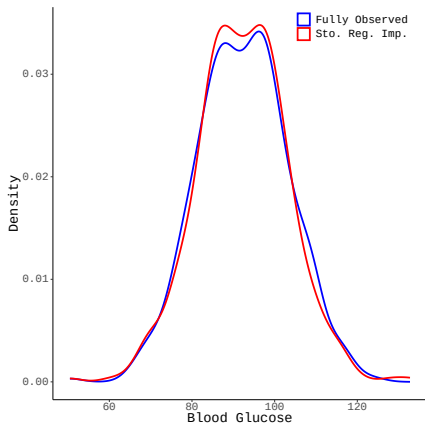
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Example



Implementation

```
miceOut <- mice(data = dat1, m = 1, seed = 42, method = "norm.nob")  
impData <- complete(1)
```



OK Methods (These sometimes work)

Nonresponse Weighting

- Weight the observed cases to correct for nonresponse bias
 - Popular in survey research and official statistics
 - Only worth considering with *Unit Nonresponse*
 - Doesn't make any sense with *Item Nonresponse*



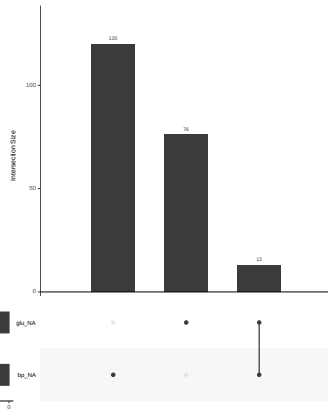
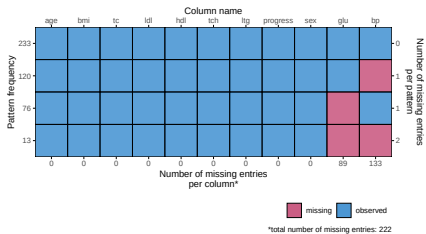
Comparison

Run a Monte Carlo simulation to compare the treatments.

- Use the synthetic diabetes data as the population.
- Simulate MAR missingness.
 - Blood Glucose
 - $PM = 20\%$, $P(M) \sim \{bmi, age\}$
 - Blood Pressure
 - $PM = 30\%$, $P(M) \sim \{bmi, tc\}$
- Treat the missing data as above.
- Use the treated data to estimate several statistics.
- Repeat the process 250 times and pool the results.



Comparison



Comparison

	MS	DRI	SRI	CC	FO
glu	91.30	92.43	92.41	91.30	92.35
bp	97.29	95.50	95.55	97.29	95.50

Variable Means

	MS	DRI	SRI	CC	FO
glu	92.98	105.69	120.70	112.26	117.29
bp	117.22	139.89	175.35	158.40	177.85

Variable Variances

Comparison

$$Y_{BP} = \beta_0 + \beta_1 X_{BMI} + \beta_2 X_{Glucose} + \beta_3 X_{Age} + \varepsilon$$

	MS	DRI	SRI	CC	FO
β_0	65.06	31.85	39.41	40.65	39.74
β_{bmi}	0.39	0.51	0.62	0.61	0.66
β_{glu}	0.16	0.44	0.32	0.32	0.30
β_{age}	0.16	0.19	0.22	0.19	0.22
R^2	0.12	0.37	0.26	0.21	0.25

Linear Regression Estimates

Comparison

	age	bmi	bp	tc	ltg
MS	29.46	15.42	25.29	92.26	1.74
DRI	44.53	21.78	66.47	118.07	2.32
SRI	43.99	21.66	61.16	118.12	2.32
CC	29.76	14.28	50.35	67.44	1.85
FO	40.52	21.19	58.19	107.83	2.28

Covariances with Blood Glucose



Comparison

	age	bmi	tc	ltg	glu
MS	33.11	12.29	34.57	1.51	25.29
DRI	54.20	22.46	92.89	2.54	66.47
SRI	54.19	22.27	90.86	2.54	61.16
CC	36.42	14.87	50.73	2.05	50.35
FO	52.50	22.65	86.00	2.60	58.19

Covariances with Blood Pressure

